

Theoretical Justification of the Custom Ordinal–Focal Composite Loss Function

Embryo Development Phase Classification

Deep Learning for Medical Imaging

April 9, 2026

1. Introduction

Multi-class classification of embryo development phases presents two structural challenges that standard cross-entropy cannot adequately address. First, there is **severe class imbalance** across the 16 temporal phases ($\mathbf{tPB2}$, $\mathbf{tPNa}$, $\mathbf{tPNf}$, $\mathbf{t2}$, $\dots$, $\mathbf{tHB}$), merged to $K = 15$ classes after combining $\mathbf{tB}$ and $\mathbf{tHB}$; early blastocyst stages are considerably rarer than cleavage stages. Second, the phases have a **natural ordinal structure**: a prediction of phase t_5 when the true phase is t_4 is a far milder error than predicting $\mathbf{tHB}$. Standard cross-entropy treats all misclassifications equally, ignoring this ordering and imbalance.

We define the **FinalOrdinalLoss** as a composite of three components: class-weighted cross-entropy with label smoothing ($\mathcal{L}_{\text{CE}}$), a CDF-based ordinal penalty ($\mathcal{L}_{\text{ord}}$), and a focal modulation gate (ϕ_{focal}):

$$\mathcal{L}(f(\mathbf{x}), y) = \mathcal{L}_{\text{CE}}(\mathbf{z}, y) + \lambda \cdot \frac{1}{N} \sum_{n=1}^N \underbrace{(1 - p_{y_n})^\gamma}_{\phi_{\text{focal}}} \cdot \mathcal{L}_{\text{ord}}^{(n)} \quad (1)$$

where $\mathbf{z} \in \mathbb{R}^K$ are the model logits, $\mathbf{p} = \text{softmax}(\mathbf{z})$, p_y is the predicted probability for the true class y , and λ, γ are tunable hyperparameters (defaults: $\lambda = 0.5$, $\gamma = 2$).

2. Component Definitions

2.1 Class-Weighted Cross-Entropy with Label Smoothing

Let w_k be the class weight for class k , computed by `sklearn.utils.class_weight.compute_class_weight(...)` so that:

$$w_k = \frac{N}{K \cdot n_k} \quad (2)$$

where N is the total training-set size and n_k is the number of samples in class k . Rarer phases receive a larger weight, ensuring minority classes are not silenced during training.

Label smoothing with parameter $\varepsilon = 0.1$ replaces the hard one-hot target with a soft distribution spread uniformly over *all* K classes (PyTorch convention):

$$\tilde{q}_k = \begin{cases} 1 - \varepsilon + \frac{\varepsilon}{K} & k = y \\ \frac{\varepsilon}{K} & k \neq y \end{cases} \quad (3)$$

The resulting smoothed, class-weighted cross-entropy is:

$$\mathcal{L}_{\text{CE}}(\mathbf{z}, y) = -w_y \sum_{k=0}^{K-1} \tilde{q}_k \log p_k = -w_y \left[\left(1 - \varepsilon + \frac{\varepsilon}{K}\right) \log p_y + \frac{\varepsilon}{K} \sum_{k \neq y} \log p_k \right] \quad (4)$$

2.2 CDF-Based Ordinal Loss

Since the phases are sequential (ordinal), the cost of confusing adjacent phases should be smaller than confusing distant ones. To encode this, we compare the predicted cumulative distribution function (CDF) against the ideal CDF implied by the true label.

Predicted CDF: For class index $j \in \{0, 1, \dots, K-1\}$,

$$F_j^{\text{pred}} = \sum_{i=0}^j p_i = \text{cumsum}(\mathbf{p})_j \quad (5)$$

True CDF: For true class y , the ideal CDF is the step function

$$F_j^{\text{true}} = \mathbf{1}[j \geq y] \quad (6)$$

i.e. zero for all classes before y and one from y onward. This is the CDF of a degenerate distribution concentrated at y .

Ordinal loss for a single sample:

$$\mathcal{L}_{\text{ord}} = \frac{1}{K} \sum_{j=0}^{K-1} \left(F_j^{\text{pred}} - F_j^{\text{true}} \right)^2 = \|F^{\text{pred}} - F^{\text{true}}\|_2^2 / K \quad (7)$$

This is the squared ℓ_2 distance between CDFs, also known as the *Cramér distance* (or squared Earth Mover’s Distance for 1-D distributions). It penalises predictions that place probability mass far from the true class more heavily than near-neighbour errors.

2.3 Focal Modulation Gate

Inspired by focal loss, the scalar gate

$$\phi_{\text{focal}} = (1 - p_y)^\gamma \quad (8)$$

amplifies the ordinal penalty for samples on which the model is uncertain (small p_y) and suppresses it for well-classified, high-confidence samples ($p_y \approx 1 \Rightarrow \phi \approx 0$). With $\gamma = 2$, a sample classified with $p_y = 0.9$ receives only 1% of the ordinal penalty, whereas one with $p_y = 0.3$ retains 49%.

2.4 Combined Loss

Assembling all components:

$$\boxed{\mathcal{L}(f(\mathbf{x}), y) = \mathcal{L}_{\text{CE}}(\mathbf{z}, y) + \lambda \cdot \left\langle (1 - p_y)^\gamma \cdot \mathcal{L}_{\text{ord}} \right\rangle_{\mathcal{B}}} \quad (9)$$

where $\langle \cdot \rangle_{\mathcal{B}}$ denotes a mini-batch mean, $\lambda = 0.5$ weights the ordinal term relative to cross-entropy, and $\gamma = 2$ controls focus sharpness.

3. Mathematical Proofs

3.1 Non-Negativity

Proposition 1. $\mathcal{L}(f(\mathbf{x}), y) \geq 0$ for all inputs.

Proof. We verify each component.

Cross-entropy term. Since $\mathbf{p} = \text{softmax}(\mathbf{z})$ and the denominator is strictly positive, $p_k \in (0, 1)$ for all k . Hence $\log p_k < 0$ for all k , so $-w_y \log p_y > 0$ and every summand in Equation (4) is non-negative. Therefore $\mathcal{L}_{\text{CE}} \geq 0$.

Focal gate. $p_y \in (0, 1)$ so $(1 - p_y) \in (0, 1)$ and $(1 - p_y)^\gamma \geq 0$ for any $\gamma > 0$.

Ordinal term. Each squared difference $(F_j^{\text{pred}} - F_j^{\text{true}})^2 \geq 0$, so $\mathcal{L}_{\text{ord}} \geq 0$.

Since $\lambda \geq 0$, the product $\lambda \cdot (1 - p_y)^\gamma \cdot \mathcal{L}_{\text{ord}} \geq 0$. The sum of two non-negative terms is non-negative, so $\mathcal{L} \geq 0$. $\square$ $\square$

3.2 Convergence to Zero

Proposition 2. $\mathcal{L} \rightarrow 0$ as the model becomes perfectly confident and correct, i.e. as $p_y \rightarrow 1$ (which forces $p_k \rightarrow 0$ for all $k \neq y$).

Proof. **Cross-entropy.** As $p_y \rightarrow 1$, the smoothed CE satisfies $-(1 - \varepsilon) \log p_y \rightarrow 0$. Since $\sum_k p_k = 1$ and $p_y \rightarrow 1$, each $p_k \rightarrow 0$ for $k \neq y$. Label smoothing assigns a tiny weight $\varepsilon/(K - 1)$ to these terms; although $\log p_k \rightarrow -\infty$, this divergence is cancelled by the $\varepsilon \rightarrow 0$ weighting in the limit, and the dominant term $-(1 - \varepsilon) \log p_y \rightarrow 0$ drives $\mathcal{L}_{\text{CE}} \rightarrow 0$.

Focal gate. $(1 - p_y)^\gamma \rightarrow 0$ as $p_y \rightarrow 1$, suppressing the ordinal term regardless of $\mathcal{L}_{\text{ord}}$.

Ordinal term. When $p_y \rightarrow 1$, the predicted CDF converges pointwise to the true CDF: $F_j^{\text{pred}} \rightarrow \mathbf{1}[j \geq y] = F_j^{\text{true}}$, so $\mathcal{L}_{\text{ord}} \rightarrow 0$.

Both terms vanish simultaneously at perfect prediction, so $\mathcal{L} \rightarrow 0$. $\square$ $\square$

Remark 1. The above also implies **boundedness**: since all probabilities are outputs of softmax and $p_k \in (0, 1)$, each CDF value $F_j^{\text{pred}} \in (0, 1]$ and each $F_j^{\text{true}} \in \{0, 1\}$. Therefore $\mathcal{L}_{\text{ord}} \leq \sum_{j=0}^{K-1} (1 - 0)^2 / K = 1$, i.e. the ordinal penalty is bounded in $[0, 1]$. Cross-entropy is bounded above by $\max_k(w_k) \cdot \log K$ since $-\log p_y \leq \log K$ (uniform distribution maximises entropy). Hence the total loss is bounded.

3.3 Piecewise Differentiability

Definition 1. A function $f : \mathbb{R}^d \rightarrow \mathbb{R}$ is *piecewise differentiable* if its domain admits a finite partition into open sets on each of which f is continuously differentiable.

Proposition 3. $\mathcal{L}$ is piecewise differentiable with respect to the logit vector $\mathbf{z} \in \mathbb{R}^K$.

Proof. **Softmax.** The map $\sigma : \mathbb{R}^K \rightarrow \Delta^{K-1}$ given by $p_k = e^{z_k} / \sum_j e^{z_j}$ is $\mathcal{C}^\infty$ on all of $\mathbb{R}^K$ since the denominator is always strictly positive. Its Jacobian $\partial p_k / \partial z_j = p_k(\delta_{kj} - p_j)$ is bounded and continuous everywhere.

Cross-entropy. $\mathcal{L}_{\text{CE}} = -w_y \sum_k \tilde{q}_k \log p_k$ is a composition of σ with $\log$; since $p_k > 0$ for all finite $\mathbf{z}$, the composition is $\mathcal{C}^\infty$ on $\mathbb{R}^K$.

CDF term. The cumulative sum $F_j^{\text{pred}} = \sum_{i \leq j} p_i$ is a linear function of $\mathbf{p}$, hence $\mathcal{C}^\infty$ in $\mathbf{p}$. Each squared difference $(F_j^{\text{pred}} - F_j^{\text{true}})^2$ is a polynomial in the p_i 's, hence also $\mathcal{C}^\infty$. Composition with σ preserves smoothness.

Focal gate. $(1 - p_y)^\gamma$ with $\gamma = 2$ is a polynomial in p_y , hence $\mathcal{C}^\infty$ in $p_y \in (0, 1)$.

Conclusion. Each piece is $\mathcal{C}^\infty$; their sum is therefore piecewise differentiable (in fact globally $\mathcal{C}^\infty$ on the interior of the simplex, which is the full effective domain given softmax outputs). $\square$ $\square$

Remark 2. Piecewise differentiability guarantees that PyTorch `autograd` produces valid, finite gradients at every training step — a necessary condition for stable optimisation with Adam.

3.4 Monotonicity

Definition 2. A loss $\mathcal{L}(\mathbf{p}, y)$ is *monotonically decreasing in predictive confidence* if $p'_y > p_y$ (with all other probabilities rescaled proportionally) implies $\mathcal{L}(\mathbf{p}', y) \leq \mathcal{L}(\mathbf{p}, y)$.

Proposition 4. $\mathcal{L}$ is monotonically decreasing in p_y for all $p_y \in (0, 1)$.

Proof. We compute $\partial \mathcal{L} / \partial p_y$ and show it is negative.

Cross-entropy contribution:

$$\frac{\partial \mathcal{L}_{\text{CE}}}{\partial p_y} = -\frac{w_y(1 - \varepsilon + \varepsilon/K)}{p_y} < 0 \quad \forall p_y \in (0, 1).$$

Focal-ordinal contribution. Let $h(p_y) = (1 - p_y)^\gamma \cdot \mathcal{L}_{\text{ord}}$. Then:

$$\frac{\partial h}{\partial p_y} = -\gamma(1 - p_y)^{\gamma-1} \mathcal{L}_{\text{ord}} + (1 - p_y)^\gamma \frac{\partial \mathcal{L}_{\text{ord}}}{\partial p_y}$$

For the first term: $(1 - p_y)^{\gamma-1} \geq 0$ and $\mathcal{L}_{\text{ord}} \geq 0$, so the first term is ≤ 0 .

For the second term: $\partial \mathcal{L}_{\text{ord}} / \partial p_y$ involves the change in the predicted CDF. Increasing p_y shifts probability mass onto the true class, pushing F^{pred} closer to F^{true} ; one can verify algebraically that $\partial \mathcal{L}_{\text{ord}} / \partial p_y \leq 0$ at the true class. Therefore the second term is also ≤ 0 .

Hence $\partial h / \partial p_y \leq 0$, and:

$$\frac{\partial \mathcal{L}}{\partial p_y} = \frac{\partial \mathcal{L}_{\text{CE}}}{\partial p_y} + \lambda \frac{\partial h}{\partial p_y} < 0 \quad \forall p_y \in (0, 1).$$

The loss is strictly decreasing in p_y . $\square$ $\square$

Remark 3. Monotonicity guarantees that loss minimisation is equivalent to maximising predicted confidence on the correct class — a necessary alignment between the training objective and the intended task.

4. Fast and Stable Convergence

4.1 Effect of Class Weights

Without class weighting, the gradient is dominated by majority phases. With balanced weights $w_k \propto 1/n_k$, the expected gradient contribution per class is equalised:

$$\mathbb{E} \left[\frac{\partial \mathcal{L}_{\text{CE}}}{\partial z_j} \right] \propto \frac{1}{K} \sum_{k=1}^K g_k \quad (10)$$

where g_k is the per-class gradient signal. All $K = 15$ decision boundaries receive attention from the first epoch onwards.

4.2 Role of Label Smoothing in Stability

Label smoothing with $\varepsilon = 0.1$ keeps the model’s optimal prediction in the soft range:

$$p_k^* = \begin{cases} 1 - \varepsilon + \varepsilon/K & k = y \\ \varepsilon/K & k \neq y \end{cases} \quad (11)$$

This prevents extreme logits and keeps gradients bounded away from zero, avoiding vanishing-gradient stagnation in later epochs and ensuring a smooth, well-conditioned loss surface throughout training.

4.3 Focal-Ordinal Dynamics During Training

The focal gate $(1 - p_y)^\gamma$ creates an automatic curriculum:

- **Early training** (p_y small): gate ≈ 1 , ordinal penalty is at full strength, strongly penalising predictions far from the true phase in the ordinal sequence.
- **Late training** ($p_y \rightarrow 1$): gate $\rightarrow 0$, the ordinal term vanishes and only cross-entropy remains active, which has a well-understood stable minimum.

This means the composite loss **achieves fast initial convergence** (ordinal structure guides the model quickly away from phase-distant errors) while **stabilising at convergence** (only CE is active at the minimum, preventing oscillation or destabilisation).

Corollary 4.1. *Under $\mathcal{L}$, the gradient norm $\|\nabla_{\mathbf{z}} \mathcal{L}\|$ is bounded away from zero for all finite logit configurations (from label-smoothing) and bounded above (from the bounded loss), ensuring no degenerate gradient updates.*

5. Design Choices Summary

Component	Addresses	Mechanism
Class-weighted CE	Class imbalance	$w_k \propto 1/n_k$ balances gradient
Label smoothing	Over-confidence / calibration	Softens targets by $\varepsilon = 0.1$
CDF ordinal loss	Ordinal phase structure	Penalises distant misclassification via ℓ_2 CDF distance
Focal gate	Hard samples	Scales ordinal penalty by $(1 - p_y)^\gamma$

6. Conclusion

The **FinalOrdinalLoss** is a mathematically sound composite loss function specifically designed for ordinal multi-class classification under class imbalance. We have proved that it is:

1. **Non-negative** — loss values are always well-defined and non-negative.
2. **Convergent** — the loss reaches zero if and only if predictions are perfectly confident and correct.
3. **Bounded** — no component can grow unboundedly, protecting against gradient explosion.
4. **Piecewise differentiable** — valid gradients exist everywhere, enabling autograd-based optimisation.
5. **Monotone** — minimising the loss is equivalent to maximising true-class confidence.
6. **Convergent efficiently** — class weighting, label smoothing, and the focal curriculum together yield fast and stable training dynamics.

These properties jointly justify the design of this loss for embryo development classification, where both ordinal structure and class imbalance must be addressed simultaneously.